

Lecture 19: Practice Problems — Question Set

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Comprehensive Case-Based & Technical Challenges

This problem set evaluates your mastery across all concepts taught in **Lecture 19 (Advanced Matplotlib & Seaborn Mastery)**:

- Continuous Distributions, Histograms, Density Normalization (`plt.hist` , `density=True`)
 - Bin Estimation Models (Sturges' Formula vs Freedman-Diaconis Rule)
 - Parametric Benchmarks & Vertical Reference Lines (`plt.axvline`)
 - Box Plots, Tukey's Five-Number Summary, and Outlier Fences ($1.5 \times \text{IQR}$)
 - Notched Box Plots for Visual Hypothesis Testing of Medians
 - Split Violin Plots and Multimodal Distribution Discovery (`sns.violinplot`)
 - Object-Oriented Subplot Grids (`fig, axes = plt.subplots`) & Scale Sharing
 - Independent Dual-Axis Affine Coordinate Systems (`ax.twinx()`)
 - Seaborn Grammar of Graphics Semantic Encodings (`hue` , `style` , `size`)
 - Correlation Matrix Heatmaps with Upper-Triangular Masking (`sns.heatmap`)
 - Edward Tufte Data-Ink Ratio Maximization
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Case 1: Algorithmic E-Commerce Logistics & Delivery Duration Distribution

 Domain: Supply Chain Analytics & Fulfillment Optimization

 Tested Topics: Lecture 19.01 – 19.03 & 19.14 (Histograms, Optimal Bins, Probability Density Normalization, Reference Lines, Gaussian KDE)

 Business Scenario:

You are the **Principal Logistics Data Scientist** at a tier-1 e-commerce platform. The operations engineering team has compiled delivery duration records across 5,000 completed orders across two distinct fulfillment mechanisms:

1. **Express Drone / Bike Courier:** Rapid urban delivery characterized by tight execution variance ($\mu = 2.5$ hours, $\sigma = 0.6$).
2. **Regional Fulfillment Warehousing:** Hub-and-spoke multi-stage delivery with right-skewed transit delays ($\mu = 7.5$ hours, $\sigma = 2.4$, heavy right tail).

The executive leadership has established a formal Service Level Agreement (SLA): **at least 90% of all platform deliveries must arrive within 6.0 hours.**

 Dataset Specification (Synthetic Generation Template):

```
import numpy as np

np.random.seed(42)
n_samples = 2500

# Express Urban Logistics (Normal Distribution)
express_durations = np.random.normal(loc=2.5, scale=0.6, size=n_samples)
express_durations = np.clip(express_durations, 0.5, 6.0)

# Regional Warehousing Logistics (Right-Skewed Lognormal Distribution)
regional_durations = np.random.lognormal(mean=1.9, sigma=0.45, size=n_samples)
regional_durations = np.clip(regional_durations, 2.0, 18.0)
```

 Problem Statements & Questions to Solve:

1. **Mathematical Bin Width Optimization:**
 - Calculate the optimal number of bins k for the regional dataset using:

■ Sturges' Rule:

$$k_{\text{sturges}} = 1 + \lceil \log_2(n) \rceil$$

- **Freedman-Diaconis Rule**:

$$h = 2 \cdot \frac{\text{IQR}(X)}{n^{1/3}} \implies k_{\text{fd}} = \left\lceil \frac{\max(X) - \min(X)}{h} \right\rceil$$

- Explain why Sturges' formula severely under-bins heavy-tailed, skewed data, whereas Freedman-Diaconis preserves tail resolution.

2. Dual-Series Normalized Density Histogram Implementation:

- Plot both delivery channels on a single canvas using `plt.hist()`.
- Set `density=True` to normalize both distributions into genuine probability densities where total integrated area equals 1.0.
- Superimpose continuous Gaussian Kernel Density Estimation (KDE) curves for both channels using `sns.kdeplot()` or `scipy.stats.gaussian_kde`.
- Use semi-transparent colors (`alpha=0.55`) and distinct edge borders (`edgecolor='black'`).

3. Parametric Benchmarks & SLA Threshold Reference Lines (`plt.axvline`):

- Add vertical dashed reference lines for:
 - Express Courier Median ($Q_{2,\text{express}}$) in dark green dashed style (`--`).
 - Regional Courier Median ($Q_{2,\text{regional}}$) in navy dashed style (`--`).
 - Target SLA Cutoff ($x = 6.0$ hours) in bold crimson dash-dot style (`-.` , linewidth 2.5).
- Add a callout annotation showing the exact percentage of regional orders breaching the 6-hour SLA boundary.

Case 2: Multi-Cohort Clinical Trial Biomarker & Outlier Audit

 Domain: Biostatistics & Oncology Clinical Trials

 Tested Topics: Lecture 19.04 – 19.07 & 19.13 (Box Plots, Five-Number Summary, Tukey's Fences, Notched Median Confidence

Intervals, Split Violin Plots)

Business Scenario:

A biopharmaceutical research group is analyzing tumor volume reduction percentage (%) in a Phase III multi-center oncology trial. A cohort of $N = 480$ patients is partitioned across 4 treatment arms ($n = 120$ patients each):

- **Cohort A (Control):** Standard Placebo (Median ≈ 6)
- **Cohort B (SOC Chemotherapy):** Conventional Cytotoxic Regimen (Median ≈ 28)
- **Cohort C (Targeted Kinase Inhibitor):** Monotherapy Small-Molecule Drug (Median ≈ 48)
- **Cohort D (Combination Immuno-Oncology):** Targeted Inhibitor + Monoclonal Antibody (Median ≈ 54)

The clinical review board requires an audit of efficacy variance, outlier identification, and a visual hypothesis test to determine if Cohort D demonstrates statistically significant superiority over Cohort C without relying on parametric Student's t-tests.

Dataset Specification (Synthetic Generation Template):

```
import numpy as np
import pandas as pd

np.random.seed(101)
n_patients = 120

cohort_a = np.random.normal(loc=6.0, scale=4.5, size=n_patients)
cohort_b = np.random.normal(loc=28.0, scale=8.0, size=n_patients)
cohort_c = np.random.normal(loc=48.0, scale=11.0, size=n_patients)

# Cohort D: Bimodal mixture (80% strong responders, 20% non-responders with genomic instability)
d_responders = np.random.normal(loc=62.0, scale=6.0, size=int(n_patients * 0.8))
d_nonresponders = np.random.normal(loc=22.0, scale=5.0, size=int(n_patients * 0.2))
cohort_d = np.concatenate([d_responders, d_nonresponders])
```

Problem Statements & Questions to Solve:

1. Tukey's Five-Number Summary & Outlier Derivation:

- For Cohort B and Cohort C, compute John Tukey's Five-Number Summary:

$$\text{Summary} = (X_{(1)}, Q_1, Q_2, Q_3, X_{(n)})$$

- Derive the lower and upper outlier fences:

$$F_L = Q_1 - 1.5 \cdot \text{IQR} \quad \text{and} \quad F_U = Q_3 + 1.5 \cdot \text{IQR}$$

- Identify any exceptional outlier patients whose tumor reduction exceeds F_U or falls below F_L .

2. Notched Box Plot Visual Hypothesis Testing:

- Construct side-by-side notched box plots (`plt.boxplot(..., notch=True)` or `sns.boxplot(..., notch=True)`).
- State the mathematical formula for the 95% confidence interval notch width:

$$\text{Notch} = Q_2 \pm 1.57 \cdot \frac{\text{IQR}}{\sqrt{n}}$$

- Based on whether the notches of Cohort C and Cohort D overlap, evaluate if their population medians differ with approximate 95% statistical confidence ($\alpha = 0.05$).

3. Multimodal Density Discovery via Violin Plots (`sns.violinplot`):

- Why does a standard box plot fail to communicate the critical clinical reality of Cohort D (the presence of a 20% resistant non-responder sub-population)?
 - Implement a horizontal or vertical Seaborn violin plot with embedded quartile markings (`inner='quartile'`) to expose the bimodal density distribution.
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Case 3: Quantitative Hedge Fund Multi-Asset Risk & Macroeconomic Regime Analysis

 Domain: Quantitative Finance & Portfolio Risk Engineering

 Tested Topics: Lecture 19.08 – 19.12 & 19.15 – 19.16 (Object-Oriented Subplots, Dual-Axis `twinx`, Heatmaps with Triangular Mask, Seaborn Relational Mapping, Edward Tufte Data-Ink)

 Business Scenario:

You are a **Quantitative Portfolio Manager** evaluating multi-asset cross-market risk across 500 trading days. The fund trades across 6 asset classes and macroeconomic indicators:

1. R_{equity} : S&P 500 Equity Index Returns (%)
2. R_{bonds} : 10-Year US Treasury Bond Returns (%)
3. VIX : CBOE Market Volatility Index (Level)
4. ΔOil : WTI Crude Oil Price Percentage Change (%)
5. π_{CPI} : Breakeven Inflation Expectations (%)
6. DXY : US Dollar Trade-Weighted Currency Index Change (%)

The portfolio risk committee requires:

- A triangular correlation matrix heatmap to detect risk clustering and multi-collinearity.
- A dual-axis time-series visualization tracking Cumulative Portfolio Returns against the VIX Volatility Index during a severe macroeconomic stress period.
- A faceted 2×2 subplot grid visualizing cross-asset dynamics engineered according to Edward Tufte's Data-Ink principles.



Dataset Specification (Synthetic Generation Template):

```

import numpy as np
import pandas as pd

np.random.seed(42)
days = 500

dates = pd.date_range("2024-01-01", periods=days, freq="B")
equity_ret = np.random.normal(0.04, 1.1, days)
bonds_ret = -0.35 * equity_ret + np.random.normal(0.01, 0.45, days)
vix = 18 + np.maximum(0, -8.0 * equity_ret + np.random.normal(0, 3.5, days))
oil_chg = 0.25 * equity_ret + np.random.normal(0.02, 1.8, days)
inflation = 2.4 + 0.15 * oil_chg + np.random.normal(0, 0.2, days)
dxy = -0.20 * equity_ret + np.random.normal(0.01, 0.5, days)

market_df = pd.DataFrame({
    "Date": dates,
    "Equity": equity_ret,
    "Bonds": bonds_ret,
    "VIX": vix,
    "Oil": oil_chg,
    "Inflation": inflation,
    "DXY": dxy
})

```

? Problem Statements & Questions to Solve:

1. Upper-Triangular Masked Correlation Heatmap (`sns.heatmap`):

- Compute the full Pearson correlation matrix $\mathbf{R} \in \mathbb{R}^{6 \times 6}$:

$$r_{jk} = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^n (x_{ik} - \bar{x}_k)^2}}$$

- Apply a triangular mask (`np.triu(np.ones_like(corr, dtype=bool))`) to eliminate redundant mirrored entries.
- Use a diverging colormap (`cmap='coolwarm'`), fix `vmin=-1.0`, `vmax=1.0`, `center=0`, format numbers with two decimal places (`fmt='.2f'`), and enable grid cell borders (`linewidths=0.75`).

2. Dual-Axis Independent Affine Time-Series (`ax.twinx()`):

- On the primary left vertical axis, plot the fund's **Cumulative Return Index** (Y_1 , base index 100.0) as a solid navy line.

- On the secondary right vertical axis, instantiate an independent coordinate system using `ax.twinx()` and plot the **VIX Volatility Index** ($\tilde{Y}_2$) with a semi-transparent crimson fill (`fill_between`).
- Eliminate visual confusion by explicitly coloring each axis spine and label to match the respective metric.

3. Object-Oriented Subplot Layout & Edward Tufte Data-Ink Optimization:

- Construct a 2×2 grid (`fig, axes = plt.subplots(2, 2, figsize=(14, 9), dpi=150)`).
- Populate the 4 panes:
 - Top-Left: Dual-Series Density Histograms (Equity vs Bond returns).
 - Top-Right: Notched Box Plot comparing factor distributions.
 - Bottom-Left: Relational Scatter Plot with Seaborn semantic hue mapping.
 - Bottom-Right: Masked Correlation Heatmap.
- Apply Edward Tufte's Data-Ink maximization rule: remove unnecessary top and right borders (`ax.spines['top'].set_visible(False)` , `ax.spines['right'].set_visible(False)`), apply subtle gridlines (`linestyle=':'` , `alpha=0.5`), and prevent label clipping using `plt.tight_layout()` .



Ready to Review Answers?

When you have implemented your code in `questions.ipynb` , compare your visual outputs and derivations against the [Complete Production Solutions](#).